Istituto di Ricerca e Certificazione per le Costruzioni Sostenibili
Notified Body No. 1994 in accordance with Regulation CPR (EU) No. 305/2011

TEST REPORT

Number:

1994-CPR-RP2549

Issuing date:

11th July 2022

Applicant:

EVEREST SHPK
BULEVARDI BLU, NR. 544
KAMZA, TIRANA, ALBANIA

Tested product:

System nodes for PVC windows, of the series with trade name "76000"

(cfr. descrizione)

Executed test:

Calculation of the thermal transmittance

Normative References:

EN 14351-1:2006+A2:2016
EN ISO 10077-2:2017

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The results in the Test Report refer exclusively to the objects tested and identified in this Test Report

1 Description of the analyzed nodes

The description and technical drawings that follow, refer to the tested samples, were declared and supplied by the applicant under hi sown responsibility.

The series undergoing testing is composed of n° 3 system nodes for PVC windows of the series with trade name “76000” supplied by EVEREST SHPK.

- Material: - PVC - rigid

- Profiles:
  - frame profile item 76103,
  - leaf profile item 76401,
  - Fixing Glass Profile item 12232,
  - Overlap Profile item 76501,
  - profile in steel item 55432,55435,

  all supplied by EVEREST SHPK, Tirana (Albania).

- Glass: three layer glazing with 43mm thickness in total

- Gaskets:
  - gasket outer in EPDM (The Gasket is produced with in PVC Profile),
  - gasket sash in EPDM (The Gasket is produced with in PVC Profile),
  - gasket central in EPDM (The Gasket is produced with in PVC Profile),

  all supplied by EVEREST SHPK, Tirana (Albania).

  - under glass insulation in PCP (polycloroprene) e.g. neoprene item XXXXX,

  all supplied by XXXXX ().

- Declared nominal dimensions: cfr. enclosed technical drawings.

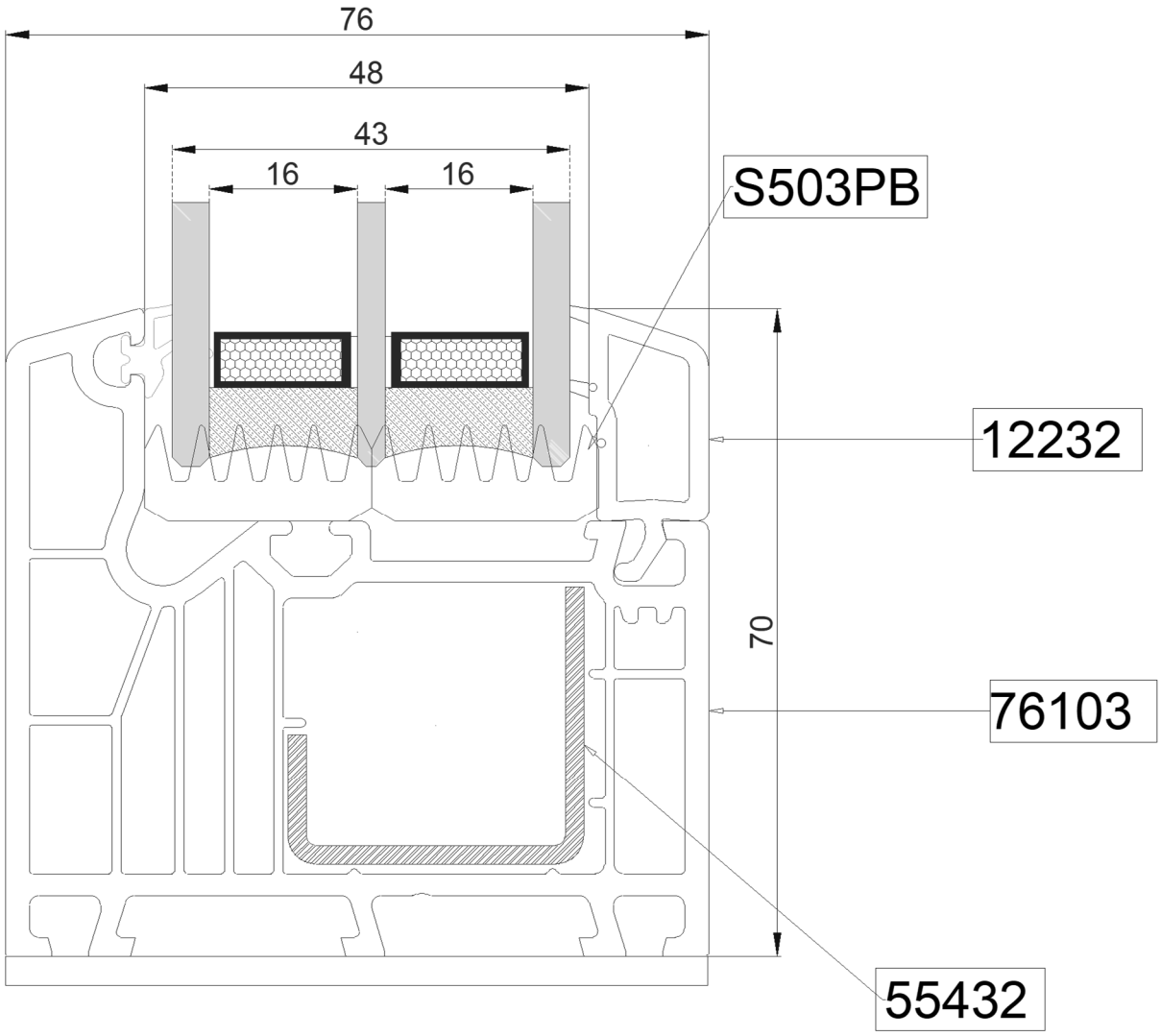


Fig. 1. Section of the node 01– Frame Sash Node
(specified nominal dimensions in mm)

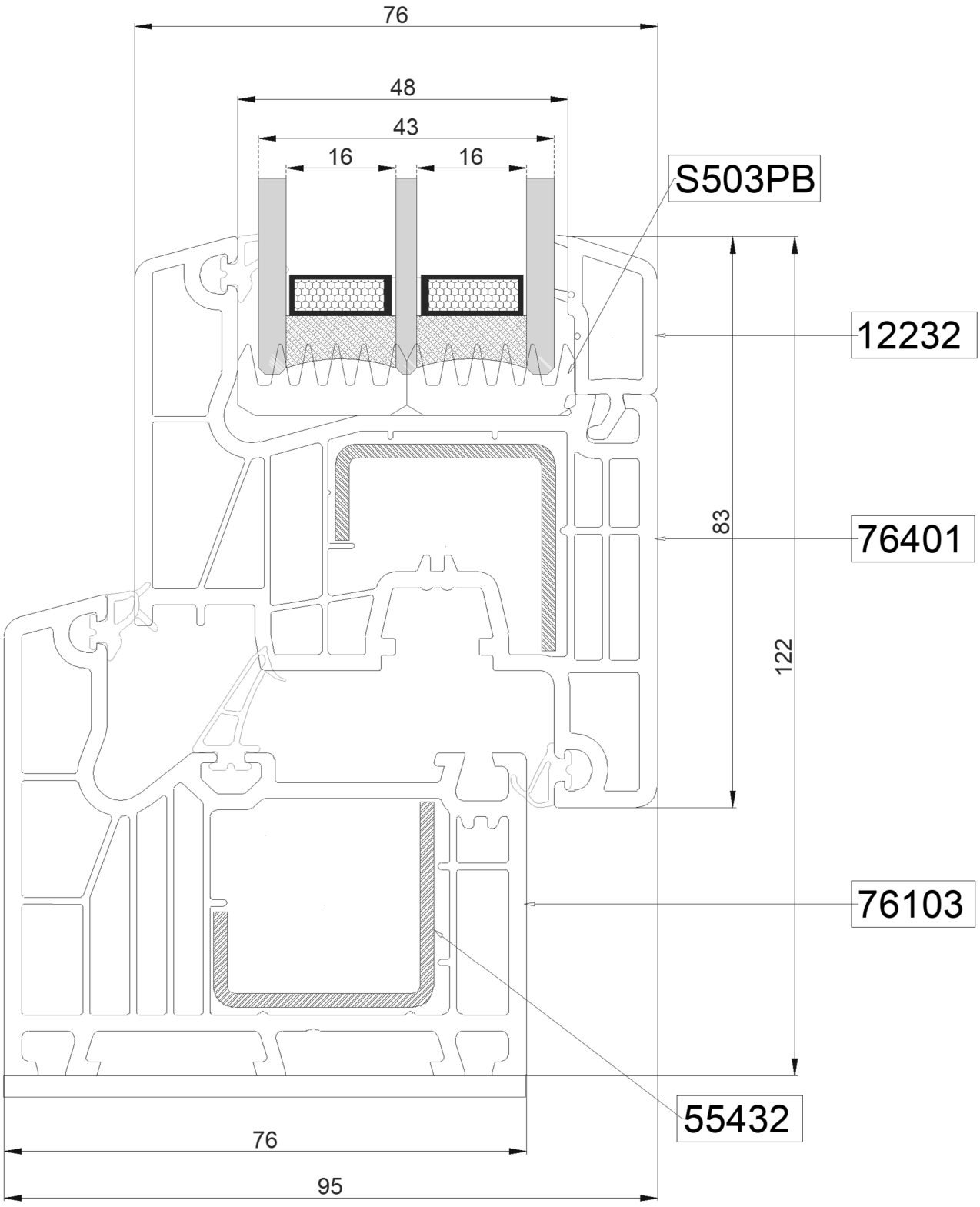


Fig. 2. Section of the node 02 – Frame Sash Node
(specified nominal dimensions in mm)

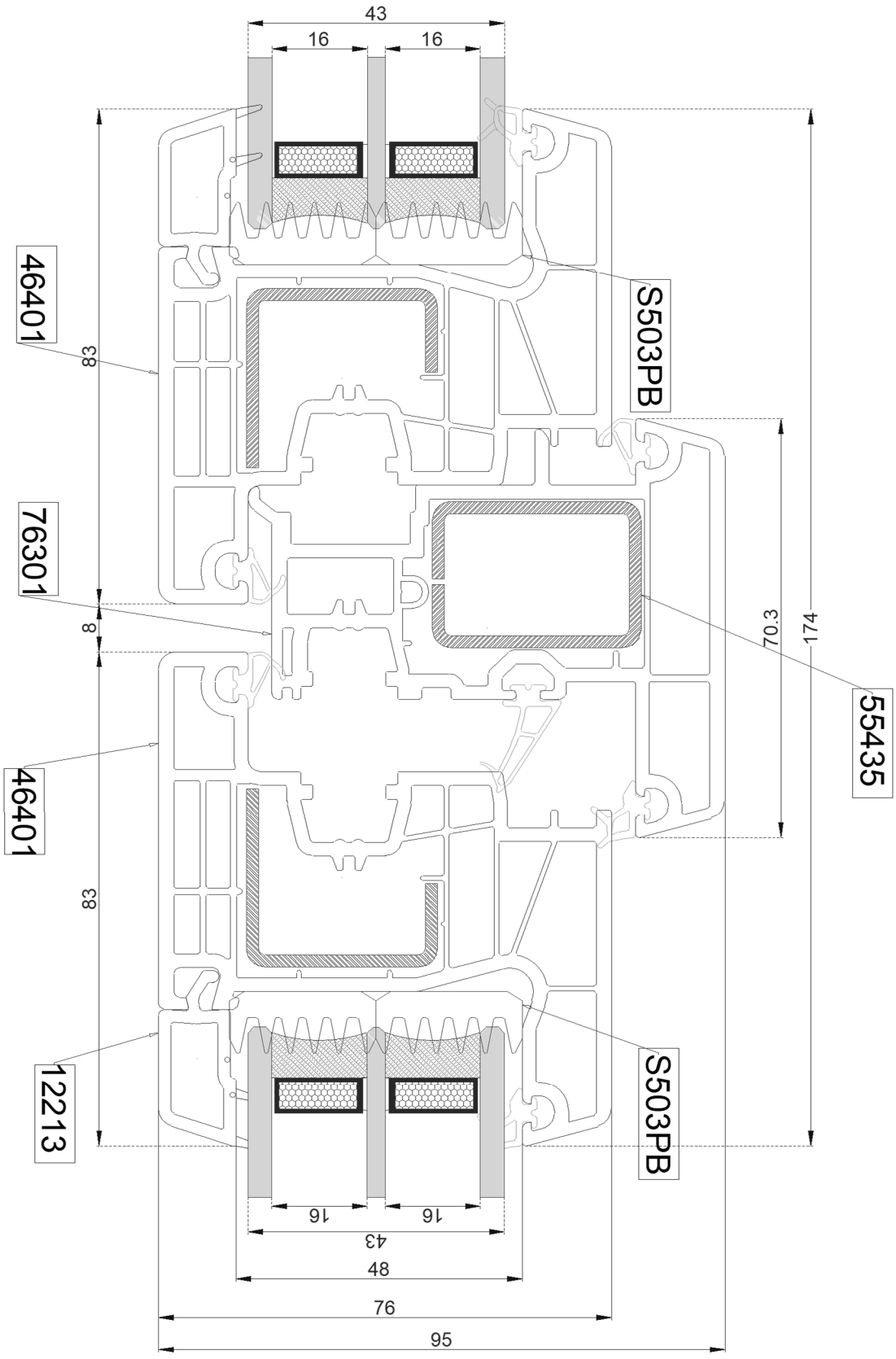


Fig. 3. Section of the node 03 – Sash Astragal Node
(specified nominal dimensions in mm)

2 Constituent materials of the analyzed sections

In Tab. 1 are reported the characteristics of the materials utilized in the analysis.

Frame Materials	Conductivity (W/mK)	Emissivity
PVC - rigid **	0,20	0,9
Steel *	50	0,9
EPDM *	0,25	0,9
PCP (polycloroprene) e.g. neoprene **	0,23	0,9

* = values derived from the EN ISO 10077-2
** = value provided by the applicant

Tab. 1. Thermal properties of the materials that compose the nodes

3 Method of analysis utilized

The thermal transmittance of frames Uf was calculated according to standard EN ISO 10077-2:2017, using the "Radiosity Method" (cf. § 6.4.2. of standard EN ISO 10077-2:2017). The "Flixo 8.1" software was used for the calculations.
Materials were chosen from software library with reference to the documentation supplied by the customer. The materials not listed in the library were added in accordance with the values supplied by the customer.

4 Result obtained

From Fig. 4 to Fig. 6 are reported the result obtained by the nodes analysis. Are reported the trend of the heat fluxes and the temperatures within the section, where each color is related to a thermal level as shown in the color legend.

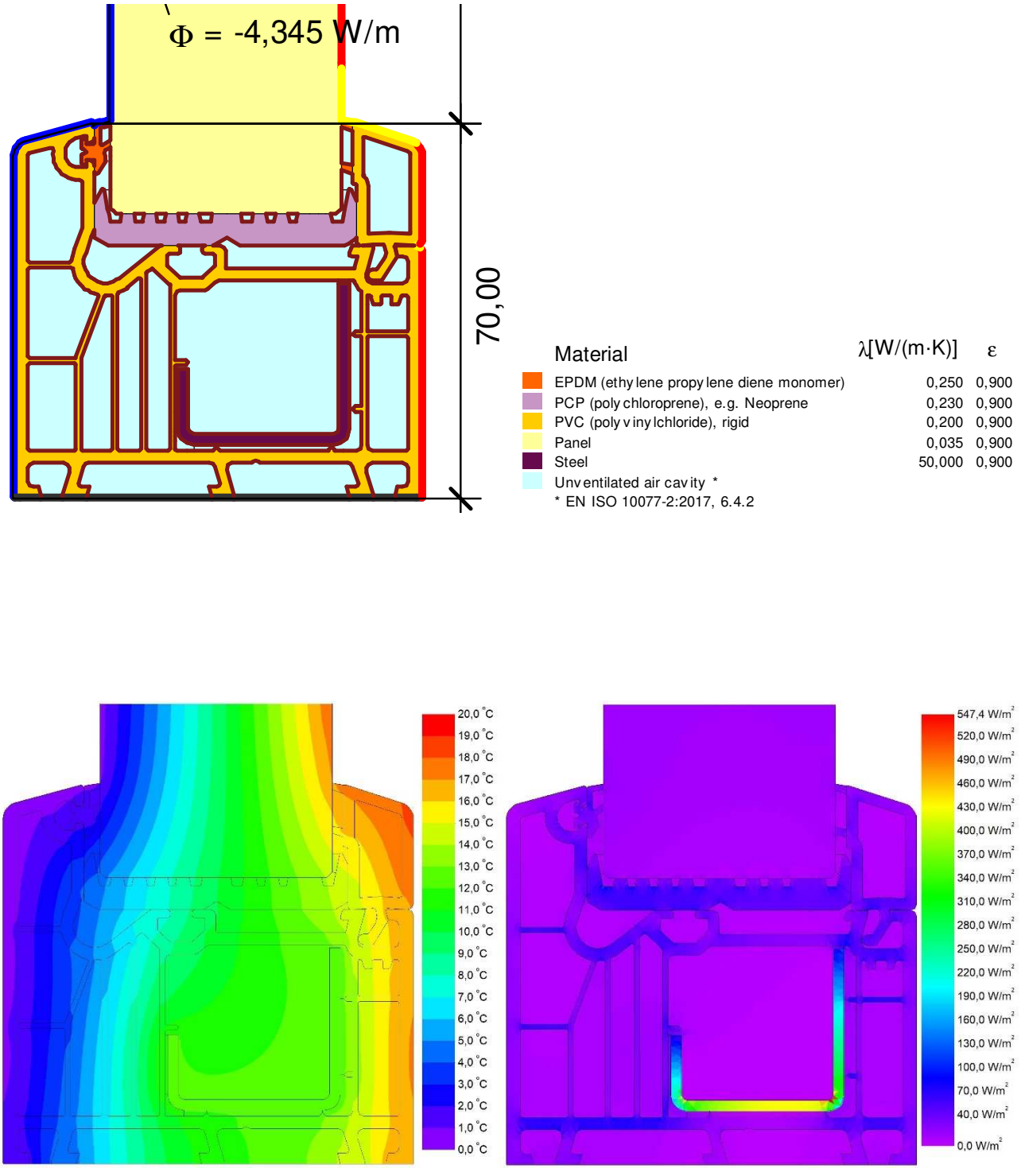


Fig. 4. Trend of temperatures and of the heat fluxes within node 01– Frame Sash Node

Thermal transmittance of node 01 – Frame Sash Node:
$U_f = 1{,}2$ W/m^{2}K

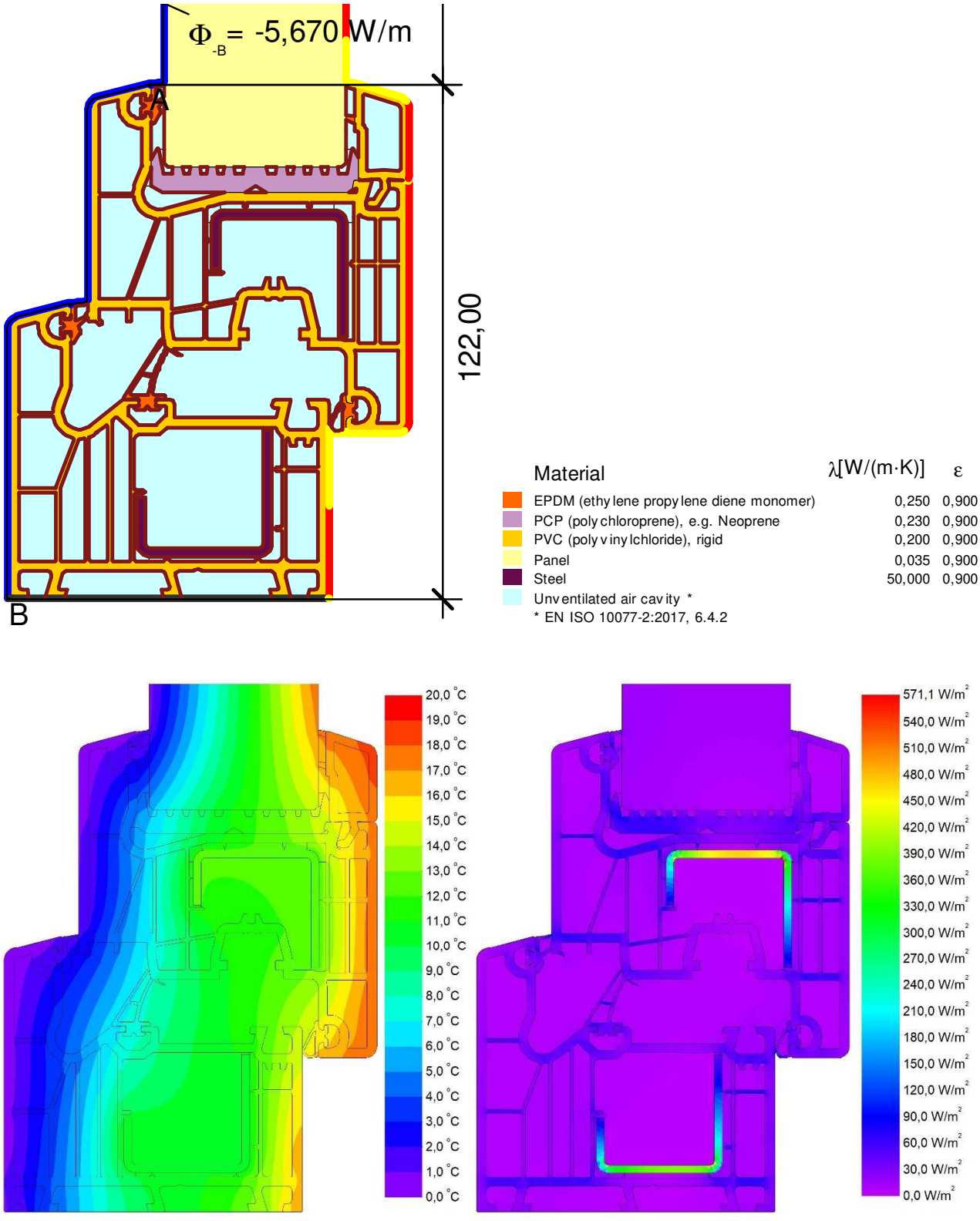
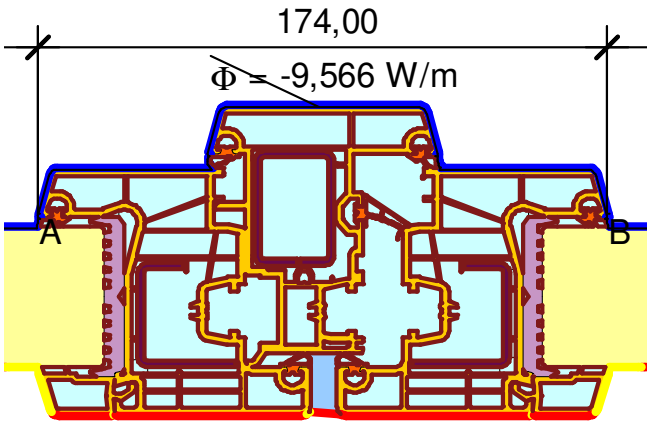


Fig. 5. Trend of temperatures and of the heat fluxes within node 02 – Frame Sash Node

Thermal transmittance of node 02 – Frame Sash Node:
U_f = 1,2 W/m^{2}K



Material	λ[W/(m·K)]	ε
EPDM (ethylene propylene diene monomer)	0,250	0,900
PCP (polychloroprene), e.g. Neoprene	0,230	0,900
PVC (polyvinylchloride), rigid	0,200	0,900
Panel	0,035	0,900
Steel	50,000	0,900
Slightly ventilated air cavity *		
Unventilated air cavity *		

* EN ISO 10077-2:2017, 6.4.2

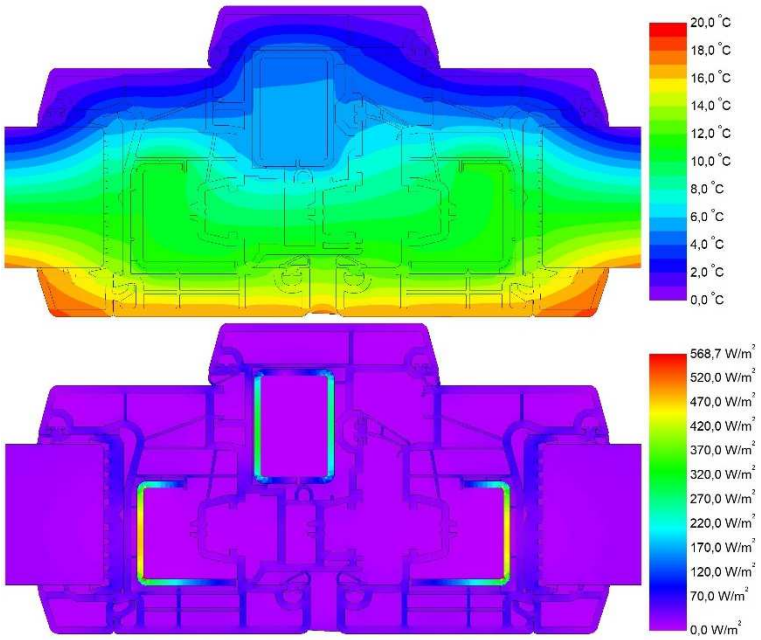


Fig. 6. Trend of temperatures and of the heat fluxes within node 03 – Sash Astragal Node

Thermal transmittance of node 03 – Sash Astragal Node:

$U_f = 1{,}2\ W/m^2K$

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